

Econometrics

Dr. Mutua Samuel

Week 7 Notes

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1 Instrumental Variables

Consider the simple regression model

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

where the error term u_i is correlated with the regressor X_i such that OLS is inconsistent for the true β_1 .

In the most simple case IV regression uses a single instrumental variable Z to obtain a consistent estimator for β_1

Regressors that are correlated with error term are called endogenous likewise that are not are called exogenous.

The remedy for this violation of the linear regression assumption is the use of instrumental variables.

Instrumental variables are variables that do not directly influence the response but are correlated with the endogenous regressor in question.

Z must satisfy two conditions to be a valid instrument:

- Instrument relevance condition: X and its instrument Z must be correlated i.e $\rho_{Z_i, X_i} \neq 0$
- Instrument exogeneity condition: The instrument Z must not be correlated with the error term u , $\rho_{Z_i, u_i} = 0$

1.1 The Two Stage Least Squares Estimator

In the first stage, the variation in the endogenous regressor X is decomposed into a problem-free component that is explained by the instrument Z and a problematic component that is correlated with the error.

The second stage uses the problem-free component of the variation in X to estimate β_1

The first stage regression model is:

$$X_i = \pi_0 + \pi_1 Z_i + v_i$$

where $\pi_0 + \pi_1 Z_i$ is the component of X_i that is explained by Z_i while v_i is the component that cannot be explained by Z_i and exhibits correlation with u_i

Using the OLS estimates $\hat{\pi}_0$ and $\hat{\pi}_1$ we obtain the predicted values $\hat{X}_i$.

If Z is a valid instrument the $\hat{X}_i$ are problem-free in the sense that $\hat{X}$ is exogenous in a regression of Y on $\hat{X}$ which is done in the second stage regression.

The second stage produces $\hat{\beta}_0^{TSLs}$ and $\hat{\beta}_1^{TSLs}$ the TSLs estimates for β_0 and β_1

For the single case instrumental one can show that the TSLs estimator of β_1 is

$$\hat{\beta}_1^{TSLs} = \frac{S_{ZY}}{S_{ZX}} = \frac{\frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})(Z_i - \bar{Z})}{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Z_i - \bar{Z})}$$

which is nothing but the ratio of the sample covariance between Z and Y to the sample covariance between Z and X.

1.2 Example

The relation between the demand for and the price of commodities is a simple yet widespread problem in economics.

Health economics is concerned with the study of how health-affecting behavior of individuals is influenced by the health-care system and regulation policy.

Probably the most prominent example in public policy debates is smoking as it is related to many illnesses and negative externalities.

We use the data set `CigarettesSW` which comes with the package `AER`. It is a panel data set that contains observations on cigarette consumption and several economic indicators for all 48 continental federal states of the U.S. from 1985 to 1995.

We are interested in estimating β_1 in

$$\log(Q_i^{cigarettes}) = \beta_0 + \beta_1 \log(P_i^{cigarettes}) + u_i \quad (1)$$

where $Q_i^{cigarettes}$ is the number of cigarettes packs per capita sold and $P_i^{cigarettes}$ is the after-tax average real price per pack cigarettes sold in state i

The instrumental variable we are going to use for instrumenting the endogenous variable is $P_i^{cigarettes}$ is `salestax` the portion of taxes on cigarettes arising from the general sales tax is measured in dollars per pack.

The idea is that `salestax` is a relevant instrument as it is included in the after-tax average price per pack.

Also, it is plausible that `salestax` is exogenous since the sales tax does not influence quantity sold directly but indirectly through the price.

```
# load the data set and get an overview
library(AER)
data("CigarettesSW")
```

We do the transformations

```
# compute real per capita prices

CigarettesSW$price <- with(CigarettesSW, price / cpi)

# compute the sales tax

CigarettesSW$salestax <- with(CigarettesSW, (taxs - tax) / cpi)
```

```
# check the correlation between sales tax and price
```

```
cor(CigarettesSW$salestax, CigarettesSW$price)
```

```
## [1] 0.6141228
```

```
# generate a subset for the year 1995
```

```
c1995 <- subset(CigarettesSW, year == "1995")
```

The first stage regression is given by:

$$\log(P_i^{cigarettes}) = \pi_0 + \pi_1 Salestax + v_i$$

```
# perform the first stage regression
```

```
cig_s1 <- lm(log(rprice) ~ salestax, data = c1995)
```

```
coeftest(cig_s1, vcov = vcovHC, type = "HC1")
```

```
##
```

```
## t test of coefficients:
```

```
##
```

```
##           Estimate Std. Error  t value  Pr(>|t|)
```

```
## (Intercept) 4.6165463  0.0289177 159.6444 < 2.2e-16 ***
```

```
## salestax    0.0307289  0.0048354   6.3549 8.489e-08 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We next store $\log(\hat{P}_i)$ the fitted values by the first stage regression

```
# store the predicted values
```

```
lcigp_pred <- cig_s1$fitted.values
```

We run the second stage regression as

```
# run the stage 2 regression
```

```
cig_s2 <- lm(log(c1995$packs) ~ lcigp_pred)
```

```
coeftest(cig_s2, vcov = vcovHC)
```

```
##
```

```
## t test of coefficients:
```

```
##
```

```
##           Estimate Std. Error t value  Pr(>|t|)
```

```
## (Intercept)  9.71988    1.70304  5.7074 7.932e-07 ***
```

```
## lcigp_pred  -1.08359    0.35563 -3.0469 0.003822 **
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The function `ivreg()` from the package `AER` carries out TSLS procedure automatically. It is used similarly as `lm()`. Instruments can be added to the usual specification of the regression formula using a vertical bar separating the model equation from the instruments.

```
# perform TSLS using 'ivreg()'
```

```
cig_ivreg <- ivreg(log(packes) ~ log(rprice) | salestax, data = c1995)
```

```
coeftest(cig_ivreg, vcov = vcovHC, type = "HC1")
```

```
##
```

```
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  9.71988    1.52832  6.3598 8.346e-08 ***
## log(rprice) -1.08359    0.31892 -3.3977 0.001411 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```